



Classification Evaluation Metrics Cheat Sheet

1. Accuracy

Definition:

Proportion of all predictions that were correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Use When:

- Classes are balanced.
- Both false positives and false negatives matter equally.

2. Precision

Definition:

Proportion of predicted positives that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Use When:

- False positives are costly (e.g., predicting a disease when absent).
- You need high certainty in positive predictions.

3. Recall (Sensitivity, True Positive Rate)

Definition:

Proportion of actual positives that were correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Use When:

- False negatives are costly (e.g., missing a disease case).
- Finding all positives is critical.

4. F1 Score

Definition:

Harmonic mean of precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Use When:

- Classes are imbalanced.
- You want a balance between precision and recall.
- One metric to track for model selection or reporting.

5. ROC AUC (Receiver Operating Characteristic Area Under Curve)

Definition:

Area under the ROC curve, which plots True Positive Rate vs. False Positive Rate across all thresholds.

Use When:

- Classes are balanced.
- You want to measure overall discrimination between classes, regardless of threshold.

6. PR AUC (Precision-Recall Area Under Curve)

Definition:

Area under the Precision-Recall curve, which plots precision vs. recall across all thresholds.

Use When:

- Classes are imbalanced (positives are rare).
- Positive class is more important.
- You care about performance across all possible thresholds.

7. Log Loss (Binary Cross-Entropy)

Definition:

Measures the average negative log likelihood of predicted probabilities.

$$\text{Log loss} = \frac{1}{N} \sum_{i=1}^N -(y_i \log(p_i) + (1 - y_i) \log(1 - p_i))$$

Use When:

- You want to evaluate the quality of probability predictions.
- Probability calibration is important.
- Both categorical accuracy and how confident the model is matters.

8. Confusion Matrix

Definition:

A table showing TP, TN, FP, FN for each class.

Use When:

- You want a detailed breakdown of model errors.
- Helpful for multi-class and imbalanced problems.

9. Specificity (True Negative Rate)

Definition:

Proportion of actual negatives that were predicted correctly.

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

Use When:

- False positives are important to control.
- Sometimes paired with recall when both positive and negative classes are critical.

How To Choose Which To Use?

- **Balanced Classes:**
 - Use **accuracy**, **ROC AUC**.
- **Imbalanced Classes:**
 - Use **precision**, **recall**, **F1 score**, **PR AUC**.
- **Negative/Positive Error Costs Differ:**
 - Use **precision** (if FP costly) or **recall** (if FN costly).
 - Consider adjusting your decision threshold.
- **Need Probability Quality:**
 - Use **log loss**.
- **Detailed Analysis:**
 - Use **confusion matrix**.

Quick Reference Table

Metric	Definition	When To Use
Accuracy	Overall correctness	Balanced data
Precision	$TP / (TP + FP)$	FP costly, high confidence needed
Recall	$TP / (TP + FN)$	FN costly, catch all positives
F1 Score	Harmonic mean	Imbalanced, balance precision/recall
ROC AUC	Area under ROC curve	Balanced, overall discrimination
PR AUC	Area under PR curve	Imbalanced, rare positives matter
Log Loss	Prob quality	Calibrated probabilities, risk estimation
Confusion Matrix	TP/TN/FP/FN details	Detailed error breakdown
Specificity	$TN / (TN + FP)$	When TN matters, paired with recall

Tip:

Report multiple metrics for a full picture—especially on imbalanced data. Tailor your choice to your application’s real-world needs, and use threshold tuning when cost of errors is unequal!

